Response to Reviewers

We thank the reviewers for their careful and constructive reports. The revision substantially sharpens the positioning of the paper, aligns the baselines, adds statistical uncertainty and heterogeneity analyses, and improves the presentation. Below we respond point by point. Section, table, and figure numbers refer to the revised manuscript.

Summary of major changes

1. **Positioning.** The manuscript is now explicitly framed as a benchmark and reproducibility contribution, not a new architecture. The Introduction states that residual forecasting and persistence blending are known techniques and that the novelty is the benchmark protocol plus the empirical characterization of failure modes. Section 5.1 adds a concrete protocol for future models.
2. **Evaluation alignment.** The ridge baseline is now evaluated on exactly the same anchors as the GRU (Section 4.3). We also quantify the effect of the training/test stride mismatch (Section 4.6) and report bootstrap confidence intervals for mean and median differences (Sections 4.2 and 4.4).
3. **Coverage and heterogeneity.** A state-level table (Table 5, Section 4.7) shows that gains are not uniform across California, Nebraska, and Texas. Per-site win/loss counts and delta-NSE quantiles are reported in Sections 4.2 and 4.4.
4. **Outlier analysis.** A new figure (Figure 7, Section 4.8) shows the distribution-shift well, its seasonal pumping-like spikes, and the failure of the calibrated model in the available test period. We explicitly state that pumping, drought, and sensor/datum changes cannot be separated without pumping records.
5. **Presentation.** The abstract and conclusions now state that mean NSE and mean KGE confidence intervals include zero and that the median NSE gain is the robust result. Ligatures were disabled (microtype) so that words such as "difficult" and "different" extract correctly. Figure 4 now uses density hexbin panels. Table 1 distinguishes the full, screened-core, and clean-core panels, and the main text states that the clean core is the primary set.
6. **References.** We added two recent groundwater ML references (Khan et al., 2023; Dey and Singh, 2025) and verified all DOIs against Crossref.

Reviewer 1

1. Limited novelty relative to journal expectations. We agree that the main contribution is not a new algorithm. The revision therefore (i) states this explicitly in the Introduction, (ii) adds Section 5.1 with a recommended evaluation protocol for future models, and (iii) positions the paper as a community benchmark resource. We also added state-level and per-site heterogeneity analyses to make the empirical contribution more substantial.

2. Limited benchmark coverage. We acknowledge the three-state, single-parameter scope and now state it more prominently in the Limitations. To partially address it, Table 5 and Section 4.7 report performance separately for California (32 wells), Nebraska (37/36 wells at 30/90 days), and Texas (5 wells). The results show that the aggregate finding is not driven by a single state, but also that gains are heterogeneous. Cross-site transfer and climate covariates are explicitly discussed in Sections 5.2 and 5.3 and left for future work.

3. Evaluation design. - *Anchor protocol:* Section 4.6 quantifies the effect of the stride mismatch. Re-evaluating at stride 3 changes mean NSE from 0.575 to 0.606 at 30 days and from -0.189 to

-0.197 at 90 days; median and KGE differences are below 0.01. - *Ridge alignment:* The ridge baseline is now evaluated on exactly the same anchors as the GRU (Section 4.3). Its median NSE is 0.780 at 30 days and 0.304 at 90 days; its mean NSE is 0.264 and -1.688, respectively. - *90-day outlier:* We now report bootstrap intervals for mean and median differences (Section 4.4) and state that the median NSE gain is the robust result. We also report the outlier well's forecasts in Figure 7 and discuss the ambiguity of the physical cause. - *Win rates:* Section 4.4 reports 55/73 positive, 13 negative, and 5 zero delta-NSE wells, with the median delta +0.115 (95% CI 0.033-0.207) and the mean +0.029 (95% CI -0.431-0.299). The state table provides further heterogeneity.

4. Statistical reporting. We added paired bootstrap confidence intervals for mean and median delta-NSE and delta-KGE at 30 and 90 days (Sections 4.2 and 4.4). We retained three seeds because of the GPU budget but now report seed variance, per-site distributions, and confidence intervals; additional seeds are a natural extension.

5. Presentation and writing. The abstract and conclusions were rewritten to avoid overemphasizing the 90-day point estimates. Figure 4 now uses density hexbins; Figure 7 adds the outlier case. Ligatures were disabled, and the manuscript was recompiled and checked for text-extraction artifacts. Table 1 and the Data section now make the hierarchy between the 279-well full panel, the 89-well screened core, and the 74-well clean core explicit.

Specific comments. - Table 1 now labels the panels "Full", "Core (screened)", and "Clean core (main)", and the caption states that the clean core is the primary analysis set. - Equation (1) now explains why site-wise z-scoring is preferred to a global scale. - The missing-data table caption and the first sentence of Section 4.6 state that the results use a stride-7 subset and are not directly comparable to Table 3. - The Zenodo DOI is displayed as `10.5281/zenodo.23749304` without a line break. - Two recent references were added and all DOIs were checked.

Reviewer 2

1. Site embeddings and unseen wells. Section 5.2 now explicitly states that the current model is evaluated on wells seen during training, that the embedding is undefined for unseen wells, and that the present results are not a leave-well-out transfer test. A proper cross-site experiment requires static predictors (well depth, aquifer, land cover, climate) and is left for future work; the repository includes the split machinery.

2. Climate covariates. Section 5.3 now explains that the univariate model cannot represent recharge, pumping, evapotranspiration, or storage changes, that these omissions likely limit 90-day absolute skill, and that precipitation, PET, and GRACE-derived storage are the most promising next steps. We also state that the reported gains come from persistence anchoring and validation calibration, not from external drivers.

3. Outlier case. Figure 7 and Section 4.8 present the outlier well's full record and 90-day forecasts. The well shows strong seasonal spikes consistent with pumping and a multi-year decline. We state that without pumping/abstraction records we cannot attribute the shift to a specific mechanism; pumping changes, drought, and sensor or datum changes are all plausible.

4. Typography and figures. Ligatures were disabled with microtype, Figure 4 was changed to hexbin density plots, and the revised PDF was checked for legibility. Figure 7 was added at 300 dpi.

Reviewer 3

The revision addresses the same major points: explicit benchmark positioning, state-level and per-site heterogeneity, aligned ridge evaluation, stride sensitivity, bootstrap confidence intervals, cautious abstract/conclusion wording, and the outlier case. The manuscript now reports 90-day median NSE 0.267 (95% CI for the median gain 0.033-0.207) and explains why the mean is sensitive to one well. The new Section 5.1 gives a concrete protocol for future benchmark users.

Note on new experiments

No new USGS data were downloaded. All new analyses use the existing processed panels and saved predictions: the aligned ridge baseline, the stride sensitivity analysis, the state-level table, the bootstrap intervals, and the outlier figure. The three-seed experiments remain the main multi-seed evidence; additional seeds are left for future work.